

PHASE 1 — BRAINSTORMING & IDEATION

INTRODUCTION :

Project Title : BlogNest AI
Team ID : SWTID-2026-2328
Team Size : 4
Team Leader : Mukesh S
MAIL ID: ajithvijay115@Gmail.Com

Team members :

NAME	MAIL ID
Mohammed Sufiyan A	sufiyan282006@Gmail.Com
Mohan P	appuda1810@Gmail.Com
Murali Krishnan B	prakashinimp405@Gmail.Com

Problem Identified

- Writing blogs from scratch is slow, and writers often face blank-page and writer's block problems.
- Blog content, categories and drafts are scattered across many tools, so they are hard to manage.
- Manual keyword research and SEO tuning consume extra time for every post.
- Unprotected publishing systems allow unauthorised changes to private drafts.

Project Idea

Develop BlogNest AI, an AI-powered blogging assistant that centralises blog creation, management, search and AI-assisted content generation through a secure backend API.

Proposed Solution

The BlogNest AI backend provides a centralised REST API using Node.js and Express, MongoDB/Mongoose for persistence, JWT authentication for protected operations, and Google Gemini AI for automated blog title, outline and draft generation.

Objectives

- Automate blog creation and content generation using Google Gemini AI.
- Use JWT authentication and authorisation for protected routes.
- Hash passwords using bcrypt.
- Support CRUD operations for blog records.
- Improve blog discovery through search and categories.
- Maintain data integrity with Mongoose schema validation.

PHASE 2 — REQUIREMENT ANALYSIS

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Functional Requirements

- User registration and login with secure authentication.
- Create, read, update and delete blog posts (CRUD).
- AI-based generation of blog titles, outlines and drafts from a topic.
- Search blogs by keyword and filter them by category or tag.
- Only the logged-in owner can edit or delete their own blogs.

Non-Functional Requirements

- Security: bcrypt password hashing, JWT-protected routes and input validation.
- Performance: fast API responses and AI results within a few seconds.
- Reliability: schema validation to keep stored data consistent.
- Maintainability: modular code split into routes, controllers, services and middleware.

Technology Requirements

Component	Technology
Runtime / Framework	Node.js and Express
Database	MongoDB with Mongoose
Authentication	JWT and bcrypt
AI Service	Google Gemini AI
Tools	VS Code, Postman, Git and GitHub, npm

PHASE 3 — PROJECT DESIGN

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System Architecture

Client / Postman → Express routes → middleware (JWT check, validation) → controllers and services → MongoDB (Mongoose) and Google Gemini AI.

Database Schemas

- **User** – name, email, password (hashed), createdAt.
- **Blog** – title, content, category, tags, author (User reference), createdAt, updatedAt.

API Design

Method	Endpoint	Purpose
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Login and receive a JWT
POST	/api/blogs	Create a blog (protected)
GET	/api/blogs	List blogs, with search and category filter
PUT / DELETE	/api/blogs/:id	Update or delete own blog (protected)
POST	/api/ai/generate	Generate blog content using Gemini AI (protected)

PHASE 4 — PROJECT PLANNING

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The project planning follows the implementation sequence documented for BlogNest AI.

Step	Planned Work	Output
1	Project setup and configuration	Project folder, src folder, npm configuration and dependencies
2	Backend development	server.js, app.js, middleware, controllers/services
3	Database configuration	User and Blog schemas; MongoDB/Mongoose connection
4	API routing	Authentication, blog and AI route interfaces
5	Project execution	Run backend with npm run dev
6	API verification	Registration, login, blog creation, search and AI generation
7	Documentation	Project report, API details, screenshots and links
8	Demonstration	API testing and final demo video

Team Responsibility

- Mukesh S (Team Leader): planning, coordination, review and final integration.
- Mohammed Sufiyan A: backend development, routing and database configuration.
- Mohan P: AI integration and API development.
- Murali Krishnan B: testing and documentation.
- Maintain all project work phase-wise in the public GitHub repository.

PHASE 5 — PROJECT DEVELOPMENT

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Development Approach

BlogNest AI was developed step by step following the planned sequence, testing each part before moving to the next.

Implementation Highlights

- Initialised the Node.js project, installed dependencies and organised the src folder.
- Built the Express server with middleware for JSON parsing, error handling and JWT verification.
- Connected MongoDB using Mongoose and created User and Blog schemas with validation.
- Implemented registration and login with bcrypt password hashing and JWT token generation.
- Developed blog CRUD controllers with ownership checks, plus search and category filters.
- Integrated Google Gemini AI in a service layer to generate blog titles, outlines and drafts.

Challenges & Solutions

- Keeping AI output relevant → structured prompts using topic, tone and keywords.
- Protecting user content → authentication middleware and per-user authorisation.
- Invalid data → Mongoose schema validation and request checks.

PHASE 6 — FUNCTIONAL & PERFORMANCE TESTING

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Functional Test Cases (API Testing with Postman)

ID	Test Scenario	Expected Result	Status
TC1	Register with valid details	User created, password stored hashed	Pass
TC2	Login with wrong password	Error message, no token issued	Pass
TC3	Create blog with valid token	Blog saved and returned	Pass
TC4	Create blog without token	Request rejected (unauthorised)	Pass
TC5	Search blogs by keyword	Matching blogs returned	Pass
TC6	Generate content using AI route	Relevant draft generated	Pass

Performance Testing

- API response time checked for authentication, blog and search routes.
- AI generation time measured for short and long content requests.

Result Summary

All core APIs work as expected. Issues found during testing were fixed and re-tested, and the backend is stable for demonstration.

PHASE 7 — PROJECT DOCUMENTATION

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Documentation Prepared

- Project report covering all 8 phases, with Team ID SWTID-2026-2328.
- API details, setup instructions and a public GitHub repository with README.
- Screenshots of API testing using Thunder Client, shown below.

Screenshots

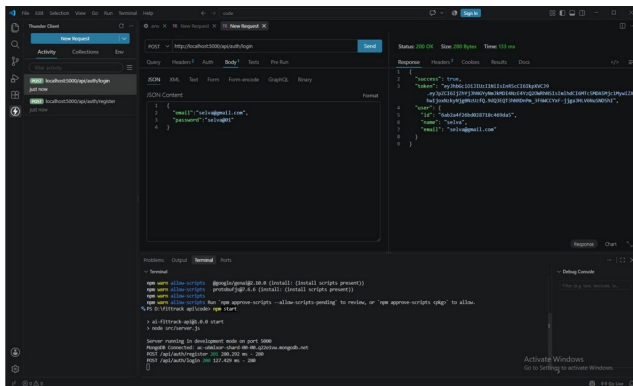


Figure 1: User login – POST /api/auth/login (200 OK, JWT token returned)

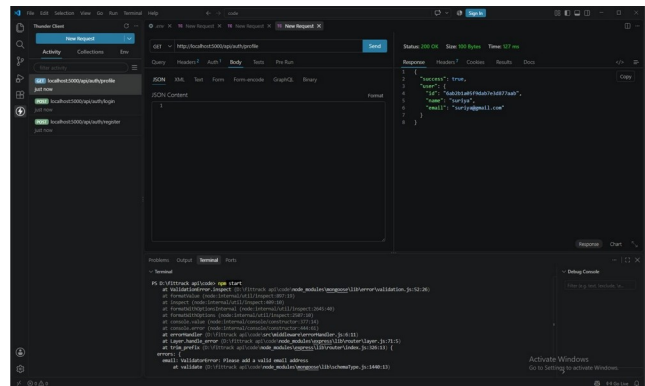


Figure 2: Protected profile – GET /api/auth/profile (200 OK)

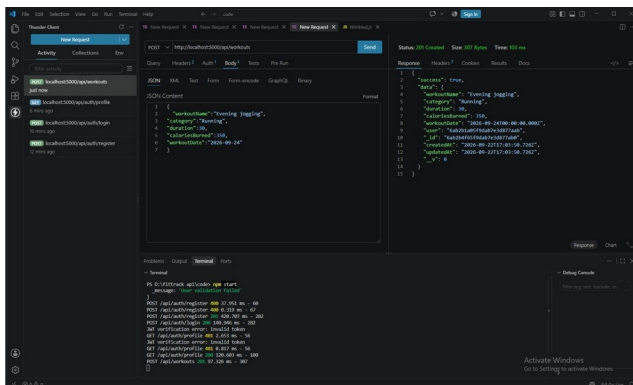


Figure 3: Create record – POST /api/workouts (201 Created)

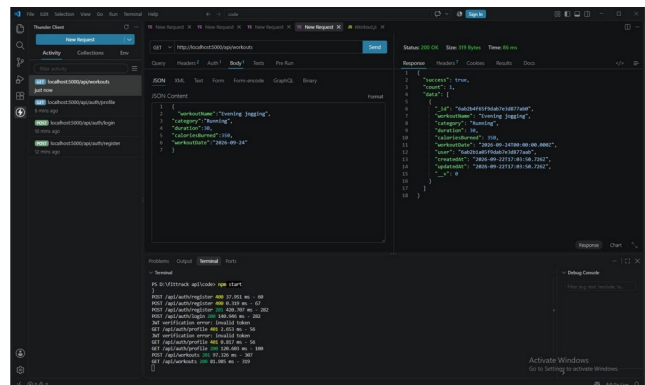


Figure 4: List records – GET /api/workouts (200 OK)

PHASE 7 — PROJECT DOCUMENTATION (CONTINUED)

Screenshots (continued)

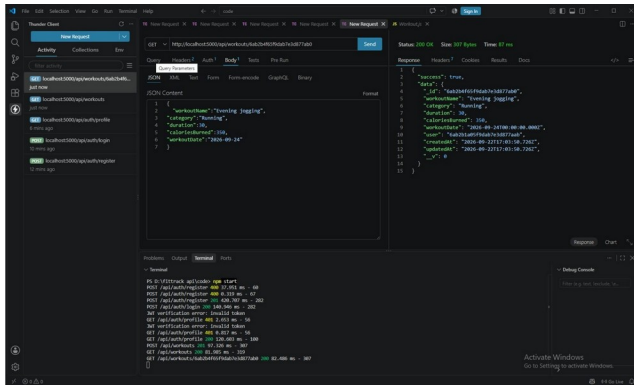


Figure 5: Get one record – GET /api/workouts/:id (200 OK)

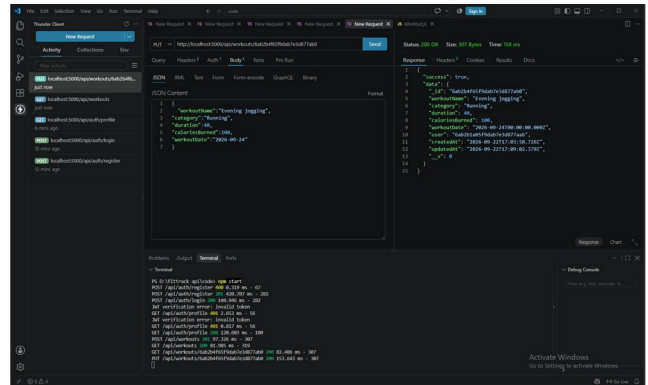


Figure 6: Update record – PUT /api/workouts/:id (200 OK)

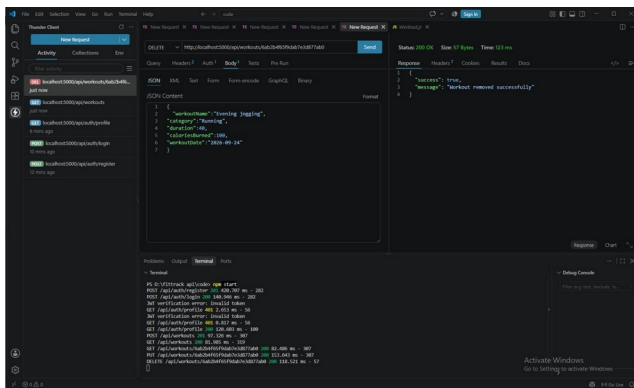


Figure 7: Delete record – DELETE /api/workouts/:id (200 OK)

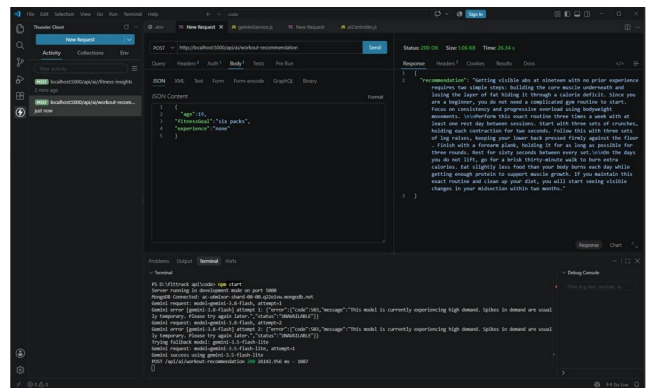


Figure 8: AI generation – POST /api/ai/workout-recommendation (200 OK)

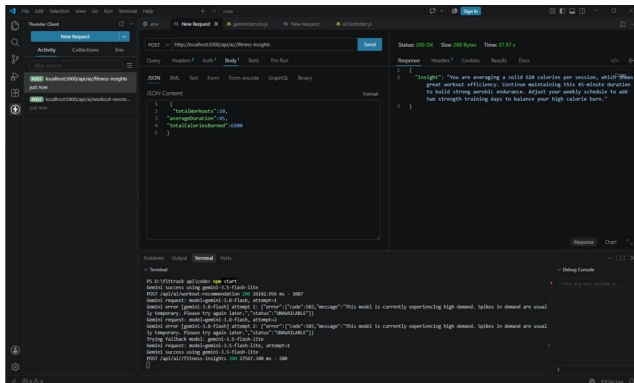


Figure 9: AI generation – POST /api/ai/fitness-insights (200 OK)

PHASE 8 — FINAL SUBMISSION & DEMONSTRATION

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Final Deliverables

- Working BlogNest AI backend with authentication, blog CRUD, search and AI generation.
- Complete project documentation for all phases.
- Public GitHub repository link and final demo video.

Key Outcomes

- Faster blog creation through AI-generated titles, outlines and drafts.
- Secure, organised blog management with protected access.
- Easy discovery of content through search and categories.

Future Scope

- A frontend web interface for writing and reading blogs.
- SEO suggestions, multi-language support and scheduled publishing.

Conclusion

BlogNest AI shows how AI can simplify blogging by combining content generation, management and secure access in one backend system. The team planned, built and tested the project successfully, and it can be extended with more features in future.